

AI Refinement of a Heath–Brown Type Zero-Free Region for the Riemann Zeta Function

<https://github.com/BlinkDL/VibeMath>

I would say this is 99% AI, although I provided some inputs and guidance.

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Abstract

Bellotti, Trudgian and Yang recently proved the explicit classical zero-free region

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.896 \log t} \right).$$

We describe a certificate-based refinement of their argument. The main new input is the explicit Fejér square

$$P_{12}(x) = \frac{|Q_{12}(e^{ix})|^2}{100000002139070},$$

where the degree-twelve polynomial Q_{12} is displayed below. Exact Bernstein-basis certificates for all prime-power orbits with $p < 100$ give

$$\sum_{p < 100} (\log p) m_p > 0.2415849,$$

and hence improve the weighted arithmetic term from $0.1186 \eta w(0)$ to

$$0.1217767 \eta w(0).$$

We then rederive the reciprocal expansion in the key Laplace-transform estimate, use a direct interval lower bound for its $m = 0$ integral, and recompute the explicit-formula error ledger. The resulting fixed-point inequality supports the constant

$$R = 4.8913.$$

More precisely, relative to the external zero-counting, gamma-factor, and verified-zero inputs already used in the cited work, the exact algebra and directed interval estimates described here imply

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8913 \log t} \right).$$

The improvement in the width $1/R$ is about 0.096%. The result is computer-assisted: the exact rational and interval computations should be independently replayed before the constant is regarded as a formally verified replacement for the published 4.896.

1 Introduction

For $s = \sigma + it$ with $\sigma > 1$, the logarithmic derivative

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}$$

turns nonnegative prime-power coefficients into a zero detector. Classical arguments combine this positivity with a nonnegative trigonometric polynomial. Stechkin's use of the functional equation

and Heath–Brown’s compactly supported smoothing function make this detector substantially stronger. Bellotti, Trudgian and Yang [1] combine these two ideas and obtain

$$\sigma > 1 - \frac{1}{4.896 \log t}, \quad t \geq 3.$$

Their proof has three numerically sensitive components:

- (i) the nonnegative trigonometric polynomial;
- (ii) a lower bound for grouped prime-power orbits;
- (iii) the finite fixed-point ledger after the explicit formula is applied.

The present note modifies all three components, but keeps the external analytic inputs of [1] unchanged: Ford’s estimate for the Laplace transform, the smoothed explicit formula, the zero-counting estimates, the verification of the Riemann hypothesis up to $3 \cdot 10^{12}$, and Yang’s explicit Littlewood region. The genuinely new computations are:

- (1) an exact degree-twelve Fejér square P_{12} ;
- (2) exact prime-power lower bounds for every prime $p < 100$;
- (3) a direct interval treatment of the dominant $m = 0$ Laplace integral;
- (4) a corrected and sharpened remainder ledger;
- (5) the final fixed-point calculation for $R = 4.8913$.

We formulate the main statement in certificate-dependent form.

Theorem 1.1 (Computer-assisted refinement). *Assume the external explicit estimates used in [1], including the verification of the Riemann hypothesis up to $H = 3 \cdot 10^{12}$ and the explicit Littlewood region*

$$\sigma > 1 - \frac{\log \log t}{21.233 \log t}.$$

If the exact rational certificates and directed interval enclosures described in Sections 4–6 are replayed, then

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8913 \log t} \right).$$

Verification status 1.2. The arithmetic certificates use exact integers and rational numbers. The remaining one-dimensional integrals and parameter inequalities use directed interval arithmetic. The note records the full mathematical reduction and the numerical margins, but it is not a proof-assistant formalization. An independent Arb, MPFI, or comparable replay is therefore desirable.

2 The smoothing function and normalization

We retain the smoothing function of [1]. Fix

$$\theta = 1.1338, \quad U = 2\theta \cot \theta,$$

and define, for $0 \leq u \leq U$,

$$w(u) = \sec^2 \theta \left(\sec^2 \theta \left(\theta \cot \theta - \frac{u}{2} \right) \cos(u \tan \theta) + 2\theta \cot \theta - u + \frac{\sin(2\theta - u \tan \theta)}{\sin 2\theta} - 2 \left(1 + \frac{\sin(\theta - u \tan \theta)}{\sin \theta} \right) \right), \quad (1)$$

with $w(u) = 0$ outside $[0, U]$. Numerically,

$$U = 1.0592329388789857 \dots, \quad w(0) = 5.6727875988616045 \dots$$

Let

$$(\kappa_0, \dots, \kappa_6) = \left(1, -\frac{851}{859}, \frac{780}{859}, -\frac{525}{859}, \frac{171}{859}, \frac{28}{859}, -\frac{29}{859} \right),$$

so that

$$\kappa := \sum_{m=0}^6 \kappa_m = \frac{433}{859}.$$

For $\delta_m = (2\sigma - 1)m$, define

$$f(u) = \eta w(\eta u) \sum_{m=0}^6 \kappa_m e^{-\delta_m u}, \quad F(z) = \int_0^\infty e^{-zu} f(u) du.$$

To avoid a harmless normalization ambiguity in displayed formulas, we set

$$f_*(0) := \eta w(0).$$

Thus the literal value of the function in (2) is $f(0) = \kappa f_*(0)$, while the explicit-formula ledger is most naturally written in units of $f_*(0)$.

Two elementary positivity facts are useful. First, if

$$p(x) = \sum_{m=0}^6 \kappa_m x^m,$$

then

$$859p(x) = \sum_{j=0}^6 b_j \binom{6}{j} x^j (1-x)^{6-j}, \quad (2)$$

where

$$(b_0, \dots, b_6) = \left(859, \frac{4303}{6}, \frac{1882}{3}, \frac{2253}{4}, \frac{7651}{15}, 469, 433 \right).$$

Hence

$$p(x) \geq \frac{433}{859} = \kappa \quad (0 \leq x \leq 1).$$

Second, w is decreasing. Indeed, after putting $v = u \tan \theta$, direct simplification gives

$$w'(u) = \sec^2 \theta \sin v \left[\tan \theta + \sec^2 \theta \left(\frac{v}{2} - \theta \right) - \tan \frac{v}{2} \right].$$

Convexity of $\tan$ on $[0, \theta]$ makes the bracket nonpositive, so $w'(u) \leq 0$.

For the final parameter range, directed interval subdivision verifies

$$f(u) \geq \kappa \eta w(0) \quad (0 \leq u \leq 59). \quad (3)$$

3 The degree-twelve Fejér square

Define

$$\begin{aligned} Q_{12}(z) = & 32635 - 54033z - 17946z^2 + 174507z^3 + 66931z^4 - 374536z^5 \\ & - 168219z^6 + 1652944z^7 + 4446322z^8 + 6169255z^9 + 5432436z^{10} \\ & + 3003044z^{11} + 837164z^{12}. \end{aligned} \quad (4)$$

Its squared Euclidean norm is

$$\mathcal{N} := \sum_{j=0}^{12} q_j^2 = 100000002139070.$$

We take

$$P_{12}(x) = \frac{|Q_{12}(e^{ix})|^2}{\mathcal{N}} = 1 + \sum_{k=1}^{12} a_k \cos(kx). \quad (5)$$

Nonnegativity is automatic. The exact numerators $\mathcal{N}a_k$ are

$$\begin{aligned} & (173779234139896, 111942599517052, 49767351006488, 12146409523234, \\ & 9367972, 9065744, 1076858103392, 588712579182, 2346122, 654528, \\ & 105539717056, 54641694280). \end{aligned}$$

Every entry is positive. The principal parameters are

$$a_1 = 1.737792304226366158 \dots, \quad (6)$$

$$a := \sum_{k=1}^{12} a_k = 3.494613602397228815 \dots, \quad (7)$$

$$P_{12}(0) = 1 + a = 4.494613602397228815 \dots \quad (8)$$

The polynomial was obtained by a constrained numerical search, followed by integer rationalization. Only the displayed integer vector is used in the proof.

4 Exact prime-power orbit certificates

For a prime p and an integer $M \geq 1$, set

$$G_{p,M}(\sigma, x) = \sum_{m=1}^M p^{-m\sigma} P_{12}(mx).$$

Since $P_{12} \geq 0$,

$$\frac{\partial}{\partial \sigma} G_{p,M}(\sigma, x) = -(\log p) \sum_{m=1}^M m p^{-m\sigma} P_{12}(mx) \leq 0. \quad (9)$$

Thus $G_{p,M}(\sigma, x) \geq G_{p,M}(1, x)$ whenever $\sigma \leq 1$.

For strict compatibility with the support range in (3), use

$$M_p = \min \left(15, \left\lfloor \frac{59}{\log p} \right\rfloor \right).$$

The infinite orbit is explicit. Writing $y = \cos x$ and denoting Chebyshev polynomials by T_k , one has

$$G_{p,\infty}(1, x) = \frac{1}{p-1} + \sum_{k=1}^{12} a_k \frac{pT_k(y) - 1}{p^2 - 2pT_k(y) + 1}. \quad (10)$$

Moreover,

$$0 \leq G_{p,\infty}(1, x) - G_{p,M_p}(1, x) \leq \frac{P_{12}(0)}{(p-1)p^{M_p}}. \quad (11)$$

For each prime $p < 100$, subtracting a rational candidate b_p , applying (11), and multiplying by the twelve positive denominators in (10) gives an integer polynomial $R_p(y)$ of degree at most 78. On replacing y by $2u - 1$, R_p is certified positive on $[0, 1]$ by exact dyadic Bernstein subdivision. The resulting lower bounds are shown in Table 1.

p	b_p	p	b_p
2	0.2383955346	43	0.0000071512
3	0.0518827925	47	0.0000054858
5	0.0075903925	53	0.0000038329
7	0.0021992359	59	0.0000027820
11	0.0004621014	61	0.0000025181
13	0.0002683487	67	0.0000019018
17	0.0001156281	71	0.0000015988
19	0.0000822567	73	0.0000014712
23	0.0000461824	79	0.0000011612
29	0.0000230998	83	0.0000010015
31	0.0000189391	89	0.0000008126
37	0.0000111868	97	0.0000006279
41	0.0000082412		

Table 1: Certified lower bounds for $G_{p,M_p}(1, x)$.

The logarithms are also bounded from below by rational arithmetic, using the positive truncation

$$\log r = 2 \sum_{j=0}^{J-1} \frac{u^{2j+1}}{2j+1} + \text{positive remainder}, \quad u = \frac{r-1}{r+1},$$

after extracting a power of two. This yields

$$\sum_{p < 100} (\log p) b_p > 0.2415849. \quad (12)$$

The high-precision value is

$$0.2415849157806932048498958094 \dots$$

Proposition 4.1 (Improved arithmetic term). *For the smoothing function in Section 2 and the polynomial P_{12} ,*

$$\sum_{n \geq 1} \frac{\Lambda(n)}{n^\sigma} f(\log n) P_{12}(t \log n) > 0.1217767 \eta w(0). \quad (13)$$

Proof. Group the terms with $n = p^m$, use (3), (9), and the orbit certificates:

$$\begin{aligned} \sum_{n \geq 1} \frac{\Lambda(n)}{n^\sigma} f(\log n) P_{12}(t \log n) &\geq \kappa \eta w(0) \sum_{p < 100} (\log p) G_{p,M_p}(1, t \log p) \\ &> \frac{433}{859} \cdot 0.2415849 \eta w(0) \\ &> 0.1217767 \eta w(0). \end{aligned}$$

□

The corresponding constant in [1] is 0.1186. Thus the arithmetic part alone improves by approximately 2.68%.

5 Recomputed analytic ledger

We now record only the changes needed in the explicit-formula calculation. The notation follows [1]. Put

$$\mu = \frac{1 - \sigma}{\eta}, \quad T_j(\sigma) = \sum_{m=0}^6 \frac{\kappa_m}{(1 + (2\sigma - 1)m)^j}.$$

Clearing denominators shows algebraically that

$$T_1'(\sigma) > 0, \quad T_2'(\sigma) > 0 \quad (\sigma \geq 1/2).$$

No floating-point root isolation is required.

The reciprocal expansion is

$$W\left(\frac{\delta_m}{\eta} + x\right) = \frac{w(0)}{\delta_m}\eta - \frac{xw(0)}{\delta_m^2}\eta^2 + E_m(\eta). \quad (14)$$

For $x = -\mu$ and $x = 1 - \mu$, the signs in the quadratic terms must be read directly from (14). Uniform estimates use $|x| \leq 1$, and Ford's parameter is scaled by η^{-1} . With these choices, the combined $m \geq 1$ terms and the $F(\sigma - \eta)$ term satisfy the following safe bound throughout the final parameter box:

$$\begin{aligned} & a_1 F(\sigma - 1 + \eta) + a_1 F(\sigma - \eta) - F(\sigma - 1) \\ & \geq I_Q(\mu) + 3.9090\eta + 20.189\eta^2 - 170\eta^3, \end{aligned} \quad (15)$$

where

$$I_Q(\mu) = \int_0^U e^{\mu u} (a_1 e^{-u} - 1) w(u) du. \quad (16)$$

The direct treatment of I_Q is important. The cubic Taylor minorant used for a coarse uniform estimate loses more than the final fixed-point margin. On the actual interval needed in Section 6, directed interval integration gives

$$I_Q(\mu) > 0.99348, \quad 0.136 < I_Q'(\mu) < 0.138. \quad (17)$$

The gamma-factor estimate gives a gain greater than $1.5630\eta w(0)$. Combining this with Proposition 4.1, the explicit-formula errors

$$53a\eta^2 + (2353 + 1601a)\eta^3,$$

and (15), one obtains

$$\frac{a\kappa w(0)}{2} \eta \log t \geq I_Q(\mu) + C_Q(\eta) - 10^{-7}, \quad (18)$$

where the safe cubic polynomial

$$C_Q(\eta) = 13.4663\eta - 165.026\eta^2 - 8117.877\eta^3 \quad (19)$$

may be used. Finally,

$$\frac{a\kappa w(0)}{2} = 4.9964370814383408 \dots < 4.99644. \quad (20)$$

For completeness, a separate interval audit of the remaining numerical lemmas gives the following safe statements in the present parameter range:

- the $t = 0$ integral occurring in the gamma error is less than 34.652; the two contributions still sum to less than $309\eta^3 < 623\eta^3$;

- for $t \geq H$, the error remains bounded by $14\eta^2 + 424\eta^3$;
- the far-zero loss remains below 10^{-8} per detector and below 10^{-7} after the trigonometric weights are summed;
- the terms $F(s_k - 1)$ with $k \geq 1$ are negligible compared with the published $10^{-10}\eta$ allowance.

6 The fixed-point calculation

Set

$$R = 4.8913, \quad A_0 = \frac{1}{R} = 0.2044446261730010\dots,$$

$$H = 3 \cdot 10^{12}, \quad T_0 = 10^{10}, \quad K = 12.$$

Define

$$\eta_0 = \frac{A_0}{\log H} = 0.007116158542429939\dots,$$

$$\sigma_0 = 1 - \frac{A_0}{\log(KH + T_0)} = 0.993450398220328925\dots$$

In particular,

$$\frac{2\sigma_0 - 1}{\eta_0} = 138.6844869400089\dots > 138,$$

so the auxiliary Ford and nonnegativity estimates remain inside their certified ranges.

The explicit Littlewood region takes over once

$$\log t \geq L_* := \exp(21.233A_0) = 76.78219271137694\dots$$

It is therefore enough to treat

$$H \leq t \leq e^{L_*}.$$

Under the usual contradiction hypothesis for the iteration lemma,

$$\eta > \frac{1}{6 \log t}.$$

Moreover, throughout this interval one may use

$$\mu \geq 1 - \frac{c_K}{\log t}, \quad c_K = \log \left(12 + \frac{1}{300} \right) + 10^{-6} = 2.485185388992674\dots$$

At the worst endpoint $\log t = L_*$ this gives

$$\mu_* = 0.9676333105211719\dots, \tag{21}$$

$$\eta_* = \frac{1}{6L_*} = 0.002170642186439818\dots \tag{22}$$

The function

$$\mathcal{B}(L) = I_Q \left(1 - \frac{c_K}{L} \right) + C_Q \left(\frac{1}{6L} \right)$$

is decreasing on $[\log H, L_*]$. Indeed,

$$L^2 \mathcal{B}'(L) = c_K I'_Q(\mu) - \frac{1}{6} C'_Q(\eta),$$

and the interval bounds in (17), together with the explicit polynomial (19), make the right-hand side strictly negative. Thus the endpoint is the only value that must be checked.

At that endpoint,

$$I_Q(\mu_*) > 0.99348, \quad C_Q(\eta_*) > 0.02836.$$

Using (18) and (20),

$$\begin{aligned} \eta \log t &> \frac{0.99348 + 0.02836 - 10^{-7}}{4.99644} \\ &> 0.2045135 \\ &> 0.2044446261730010 = A_0. \end{aligned}$$

This contradicts the assumption that a zero satisfies

$$(1 - \beta) \log t < A_0 + 10^{-100}.$$

Hence the iteration advances by 10^{-100} until the value A_0 is reached. The verified-zero computation below H and Yang's Littlewood region above e^{L^*} complete the proof of Theorem 1.1, subject to the certificate status stated after the theorem.

7 What is genuinely improved?

The improvement from 4.896 to 4.8913 is small. In terms of the width $1/R$ it is approximately

$$100 \left(\frac{4.896}{4.8913} - 1 \right) = 0.0961\%.$$

The improvement is not obtained from one isolated numerical change. It is the combined effect of:

- (i) the larger prime-power contribution $0.1217767 \eta w(0)$;
- (ii) the smaller total Fourier mass a of P_{12} ;
- (iii) direct interval evaluation of $I_Q(\mu)$ on the final μ -range;
- (iv) a rederived reciprocal expansion and a substantially smaller cubic remainder in the Laplace-transform ledger.

The prime-power improvement alone is not enough: changing the polynomial also changes the target-zero coefficient, the gamma term, and every explicit-formula error weighted by $\sum a_k$. This is why the complete fixed-point ledger is essential.

8 Reproducibility protocol

The new computation separates into exact and interval parts.

Exact arithmetic

1. Verify $\mathcal{N}$ and all autocorrelations of the integer coefficient vector in (4).
2. For each prime $p < 100$, construct the degree-78 integer polynomial $R_p(y)$ arising from (10)–(11).
3. Map $[-1, 1]$ to $[0, 1]$ and certify positivity using exact rational Bernstein subdivision.
4. Bound $\log p$ from below by a positive atanh series to prove (12) with rational arithmetic.

Directed interval arithmetic

1. Verify (3) on $[0, 59]$.
2. Enclose the integrals defining I_Q and I'_Q on $[\mu_*, 1]$.
3. Enclose the Ford remainders entering (15).
4. Check the corrected gamma-error integral and the fixed-point monotonicity.

All inequalities have visible slack at $R = 4.8913$; nevertheless, an independent implementation is recommended because the theorem is computer-assisted and the improvement is numerically modest.

9 Concluding remarks

The main arithmetic observation is that nonnegative trigonometric polynomials should not be used merely pointwise. Grouping

$$p, p^2, p^3, \dots$$

turns a weak inequality $P(x) \geq 0$ into a rigid finite-orbit lower bound. For P_{12} , exact orbit certificates recover a 2.68% improvement in the arithmetic detector while preserving nearly the same leading smoothing efficiency. The final constant improves only by about 0.096%, reflecting the compensating gamma and explicit-formula costs.

This is a refinement inside the classical zero-free-region method. It does not approach a fixed half-plane and does not constitute progress toward the Riemann hypothesis at the relevant scale. Its value is instead methodological: it gives a reproducible example in which prime-power arithmetic, exact polynomial positivity, interval analysis, and the functional-equation pairing can be jointly optimized without hiding the proof behind a single numerical black box.

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